

Neutrophil granulocytes recruited upon translocation of intestinal bacteria enhance graft-versus-host disease via tissue damage

Lukas Schwab^{1,18}, Luise Goroncy^{1,2,18}, Senthilnathan Palaniyandi^{3,4,18}, Sanjivan Gautam^{2,5,6}, Antigoni Triantafyllou⁷, Attila Mocsai⁸, Wilfried Reichardt⁹, Fridrik J Karlsson³, Sabarinath V Radhakrishnan^{3,4}, Kathrin Hanke^{1,2}, Annette Schmitt-Graeff¹⁰, Marina Freudenberg^{11,12}, Friederike D von Loewenich⁵, Philipp Wolf¹³, Franziska Leonhardt^{1,2}, Nicoleta Baxan⁹, Dietmar Pfeifer¹, Oliver Schmah¹, Anne Schönle¹, Stefan F Martin¹⁴, Roland Mertelsmann¹, Justus Duyster¹, Jürgen Finke¹, Marco Prinz^{6,15}, Philipp Henneke^{7,16}, Hans Häcker¹⁷, Gerhard C Hildebrandt^{3,4,19}, Georg Häcker^{5,6,19} & Robert Zeiser^{1,6,12,19}

Acute graft-versus-host disease (GVHD) considerably limits wider usage of allogeneic hematopoietic cell transplantation (allo-HCT). Antigen-presenting cells and T cells are populations customarily associated with GVHD pathogenesis. Of note, neutrophils are the largest human white blood cell population. The cells cleave chemokines and produce reactive oxygen species, thereby promoting T cell activation^{1,2}. Therefore, during an allogeneic immune response, neutrophils could amplify tissue damage caused by conditioning regimens. We analyzed neutrophil infiltration of the mouse ileum after allo-HCT by *in vivo* myeloperoxidase imaging and found that infiltration levels were dependent on the local microbial flora and were not detectable under germ-free conditions. Physical or genetic depletion of neutrophils reduced GVHD-related mortality. The contribution of neutrophils to GVHD severity required reactive oxygen species (ROS) because selective *Cybb* (encoding cytochrome b-245, beta polypeptide, also known as NOX2) deficiency in neutrophils impairing ROS production led to lower levels of tissue damage, GVHD-related mortality and effector phenotype T cells. Enhanced survival of Bcl-xL transgenic neutrophils increased GVHD severity. In contrast, when we transferred neutrophils lacking Toll-like receptor-2 (TLR2), TLR3, TLR4, TLR7 and TLR9, which are normally less strongly activated by translocating bacteria, into wild-type C57BL/6 mice, GVHD severity was reduced. In humans, severity of intestinal GVHD strongly correlated with levels of neutrophils present in GVHD lesions. This study describes a new potential role for neutrophils in the pathogenesis of GVHD in both mice and humans.

T cells are the main cause of tissue damage in GVHD, but the events leading up to T cell recruitment and activation are not well understood. To identify additional cell populations that may invade the gastrointestinal tract after total body irradiation (TBI), we performed gene expression analysis of gastrointestinal tracts isolated from untreated mice and mice that had received 9-Gy TBI 24 h or 48 h beforehand. Ly6G, a key neutrophil surface antigen³, was the second most differentially expressed mRNA in the gastrointestinal tract of mice after TBI (Fig. 1a,b). To test for a functional role for neutrophils in GVHD-related mortality, we used an anti-Ly6G antibody-based depletion approach³. Treatment with this antibody reduced the relative and absolute numbers of circulating neutrophils without affecting the numbers of monocytes or macrophages significantly (Fig. 1c–e) and improved survival in animals undergoing major histocompatibility complex (MHC) mismatch allo-HCT (C57BL/6 into BALB/c) compared to the IgG isotype-treated group (Fig. 1f). Levels of interferon- γ (IFN- γ) in the serum of allo-HCT recipients that received anti-Ly6G were reduced, consistent with a general reduction in inflammation (Fig. 1g). Histopathological evidence of GVHD was milder in the group receiving anti-Ly6G (Fig. 1h). This is consistent with the concept of neutrophil-induced tissue damage after allo-HCT and with results showing that SMAD3 reduces GVHD severity by restraining neutrophil function⁴. In a syngeneic transplant model, anti-Ly6G antibody had no negative impact on survival (Fig. 1i). To examine whether this neutrophil contribution was specific to the particular microbial environment present at our institution in Germany, we

¹Department of Hematology and Oncology, University Medical Center, Freiburg, Germany. ²Faculty of Biology, Albert-Ludwigs-University, Freiburg, Germany. ³Division of Hematology and Oncology, Louisiana State University Health Sciences Center Shreveport, Shreveport, Louisiana, USA. ⁴Division of Bone Marrow Transplantation, University of Utah School of Medicine, Huntsman Cancer Institute, Salt Lake City, Utah, USA. ⁵Department of Medical Microbiology and Hygiene, University Medical Center, Freiburg, Germany. ⁶Spemann Graduate School of Biology and Medicine, Albert-Ludwigs-University Freiburg, Germany. ⁷Center of Chronic Immunodeficiency, Albert-Ludwigs-University Freiburg, Germany. ⁸Department of Physiology, Semmelweis University School of Medicine, Budapest, Hungary. ⁹Department of Radiology Medical Physics, University Medical Center, Freiburg, Germany. ¹⁰Department of Pathology, University Medical Center, Albert-Ludwigs-University, Freiburg, Germany. ¹¹Max Planck Institute of Immunobiology and Epigenetics, Freiburg, Germany. ¹²Centre for Biological Signaling Studies BIOS, Albert-Ludwigs-University Freiburg, Germany. ¹³Department of Urology, University Medical Center, Freiburg, Germany. ¹⁴Allergy Research Group, Department of Dermatology, University Medical Center, University Freiburg, Germany. ¹⁵Department of Neuropathology, University Medical Center, Freiburg, Germany. ¹⁶Center for Pediatrics and Adolescent Medicine, University Medical Center, Freiburg, Germany. ¹⁷Department of Infectious Diseases, St. Jude Children's Research Hospital, Memphis, Tennessee, USA. ¹⁸These authors contributed equally to this work. ¹⁹These authors jointly supervised this work. Correspondence should be addressed to R.Z. (robert.zeiser@uniklinik-freiburg.de).

Received 21 July 2013; accepted 3 March 2014; published online 18 May 2014; doi:10.1038/nm.3517

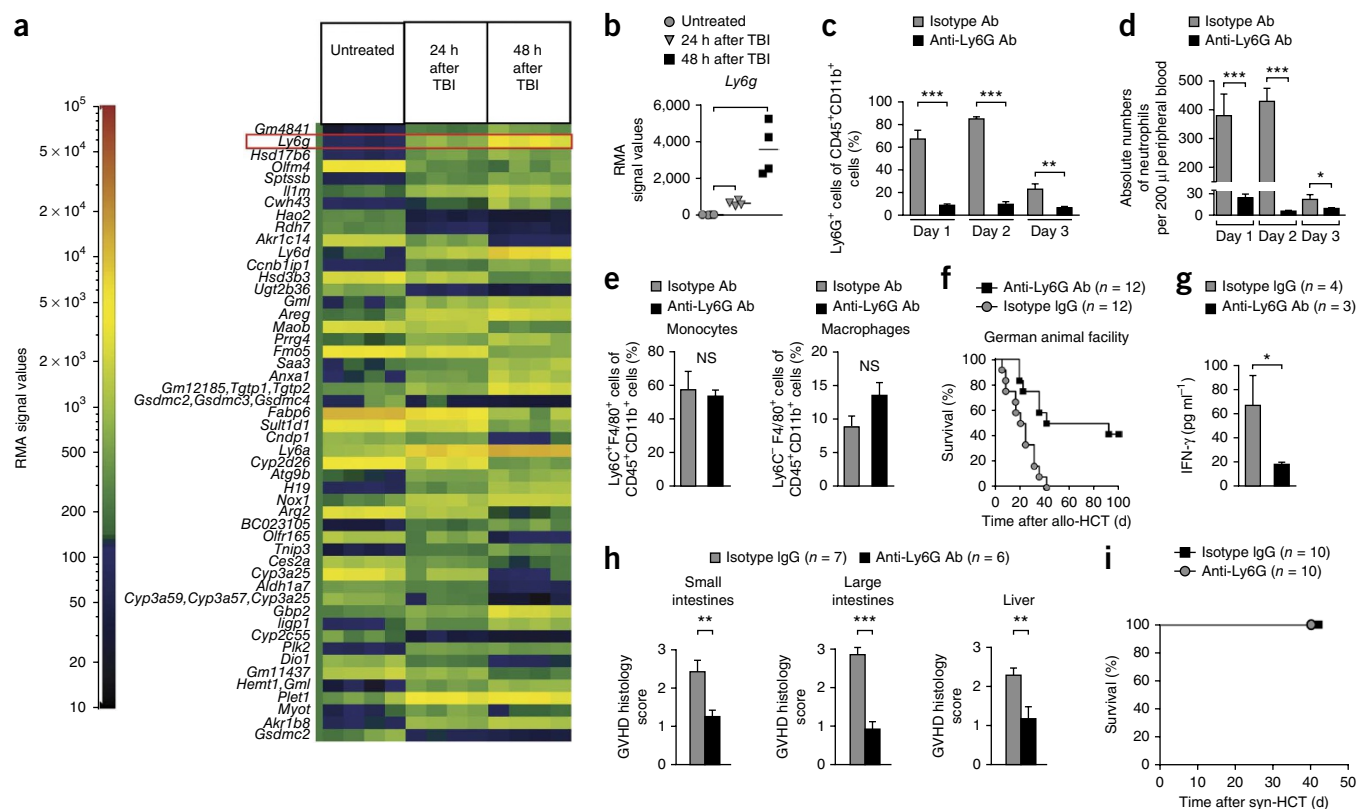


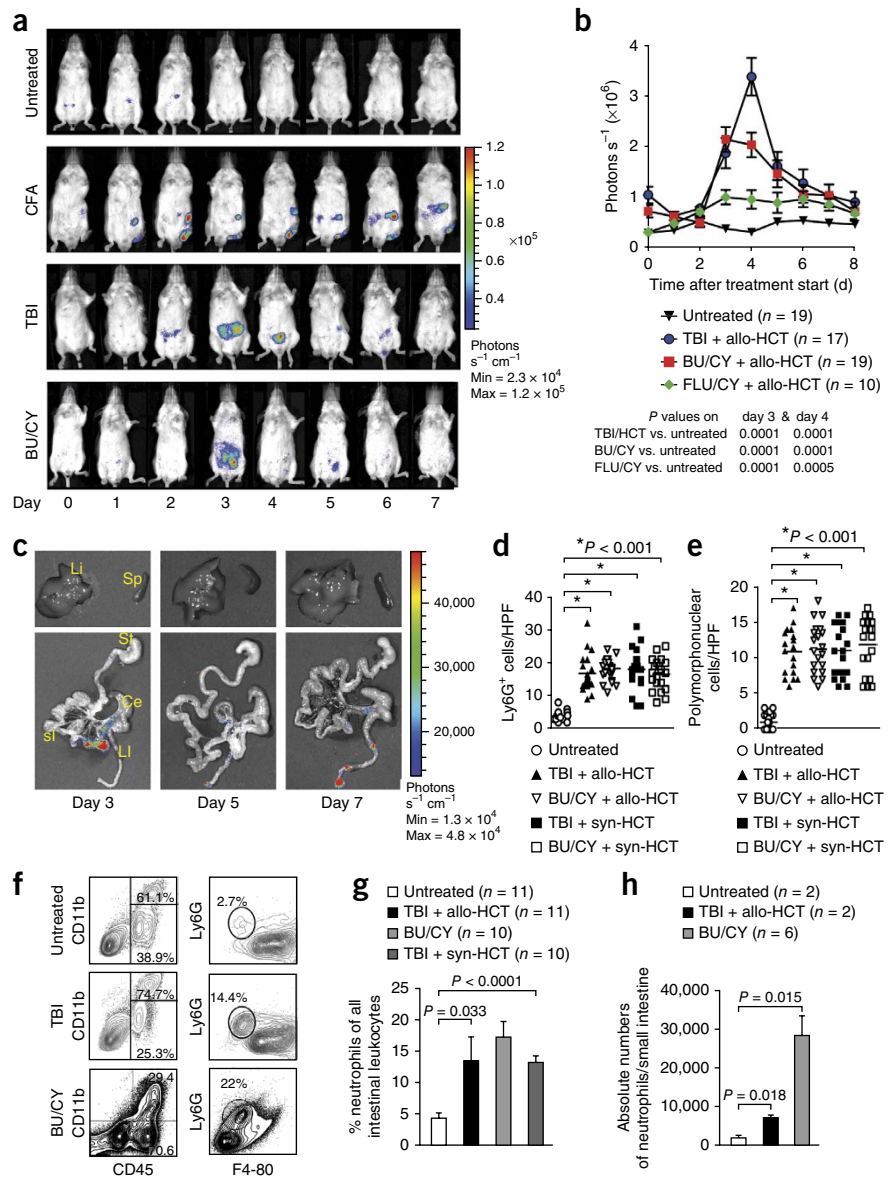
Figure 1 Anti-Ly6G antibody treatment improves survival after MHC-mismatched allo-HCT. **(a,b)** Microarray-based analysis of RNA expression levels of multiple genes in the terminal ileum of BALB/c mice after TBI (9 Gy), $n = 4$ for each group. **(a)** Tile display for the 50 most significantly regulated genes, expressed by Robust Multichip Average (RMA) signal values of 4 individual samples from the following groups: untreated, 24 h after TBI and 48 h after TBI. Red square, *Ly6g* is the second most significantly regulated gene. **(b)** Expression levels of *Ly6g* in the intestines at different time points after TBI. Each data point represents an individual animal. Untreated versus 24 h, $P < 0.0001$; untreated versus 48 h, $P < 0.0001$, Benjamini-Hochberg q value 2.675×10^{-5} . An unpaired Student's t -test (two-sided) was applied to calculate P values. All P values were calculated using a Benjamini-Hochberg multiple-testing correction. The experiment was performed once. **(c,d)** The proportion of CD45⁺CD11b⁺Ly6G⁺ neutrophils in peripheral blood at different time points after allo-HCT from C57BL/6 into BALB/c mice. Relative **(c)** and absolute **(d)** numbers of neutrophils are shown from mice injected with 500 μ g anti-Ly6G antibody or isotype control antibody per mouse on day 1 of allo-HCT. Bar diagrams represent mean $\pm$ s.e.m., $n = 6$ for each time point, with results pooled from two independent experiments. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, unpaired Student's t -test (two-sided). **(e)** The proportion of monocytes and macrophages in peripheral blood at day 2 after allo-HCT from C57BL/6 into BALB/c mice. Bar diagrams represent mean $\pm$ s.e.m., $n = 6$ for each time point (except for day 3 where $n = 7$), with results pooled from two independent experiments. NS, not significant, $P > 0.05$, unpaired Student's t -test (two-sided). **(f)** Survival of BALB/c recipients receiving allo-HCT from C57BL/6 donors ($n = 12$ per group). Data are pooled from two experiments (6 mice per experiment at Freiburg University, Germany). Isotype IgG versus anti-Ly6G, $P = 0.007$. Differences in animal survival were analyzed by log-rank test. **(g)** IFN- γ in the serum of the indicated groups on day 7 (from mice injected with 500 μ g anti-Ly6G antibody per mouse on day 1 of allo-HCT). Data (mean $\pm$ s.e.m.) are pooled from two independent experiments. $*P < 0.05$, unpaired Student's t -test (two-sided). **(h)** Scored histopathologic changes in the small intestines, large intestines and liver isolated on day 7 after allo-HCT (from mice injected with 250 μ g anti-Ly6G antibody per mouse on day 1 of allo-HCT). Data (mean $\pm$ s.e.m.) are pooled from two independent experiments. $**P < 0.01$, $***P < 0.001$, unpaired Student's t -test (two-sided). **(i)** Percentage survival of syn-HCT recipients (B6D2F1 in B6D2F1 model) injected with 500 μ g anti-Ly6G antibody or isotype control antibody on days 1 to 5 after syn-HCT. Results are representative of two experiments. **(j)** Percentage survival of BALB/c recipients receiving allo-HCT from C57BL/6 donors. Data are pooled from the three independent experiments (Shreveport, Louisiana, USA, 1 experiment; Salt Lake City, Utah, USA, 2 experiments). Isotype IgG versus anti-Ly6G, $P = 0.0014$. Differences in animal survival were analyzed by log-rank test.

also performed an experiment comparable to the one described in **Figure 1f** at an animal facility in the United States, where the mice presumably have a different microflora. We saw improvement in the survival of the US mice when they received the anti-Ly6G antibody after TBI and allo-HCT (**Fig. 1j**) or when mice (Germany) were conditioned with busulphan and cyclophosphamide (BU/CY) chemotherapy (**Supplementary Fig. 1a**) and when we used a minor histocompatibility antigen-mismatched model (BALB/B into C57BL/6, Germany) of allo-HCT (**Supplementary Fig. 1b**).

To determine the anatomical locations in which neutrophils accumulate upon allo-HCT, we used a recently described bioluminescence

method⁵ that detects a key neutrophil enzyme, active myeloperoxidase. We observed peak myeloperoxidase activity on days 3 and 4 following allo-HCT and used complete Freund's adjuvant as a positive control (**Fig. 2a,b**). Nonmyeloablative conditioning (fludarabine and cyclophosphamide) led to less myeloperoxidase signal compared to myeloablative conditioning (busulphan and cyclophosphamide) (**Fig. 2b**). To more precisely locate the myeloperoxidase signal, we performed *ex vivo* imaging of the organs from the abdominal region where we detected the signal. On day 3, the main signal came from the terminal ileum (**Fig. 2c**). Histological analysis of the terminal ileum confirmed increased infiltration of Ly6G⁺ cells (**Fig. 2d**) and

Figure 2 Myeloperoxidase-positive cells accumulate on day 3 following allo-HCT in the terminal ileum. **(a)** Bioluminescence imaging (BLI) performed 10 min after injection of luminol, in one representative BALB/c mouse from each of the following groups: untreated mice ($n = 19$), mice injected subcutaneously with complete Freund's adjuvant (CFA) into the left footpad ($n = 6$), mice that received TBI and allo-HCT ($n = 17$) and mice that received chemotherapy (BU/CY) and allo-HCT ($n = 19$). **(b)** Serial myeloperoxidase-specific imaging performed with untreated BALB/c WT mice (Untreated) and BALB/c WT mice that had undergone allo-HCT with prior conditioning with TBI (TBI + allo-HCT) or myeloablative chemotherapy (BU/CY + allo-HCT) or nonmyeloablative chemotherapy with fludarabine and cyclophosphamide (FLU/CY + allo-HCT). Allo-HCT recipients were injected with 5×10^6 BM cells and 3×10^5 T cells from C57BL/6 donors intravenously (i.v.). Data are pooled from four experiments. An unpaired Student's *t*-test (two-sided) was applied for each time point, and means $\pm$ s.e.m. are shown. **(c)** *Ex vivo* detection of active myeloperoxidase in the terminal ileum after allo-HCT by luminol BLI. Representative images of 9 individual photographs of organs from different mice at the time points day 3, day 5 and day 7. St, stomach; sl, small intestine; Ce, cecum; LI, large intestine; Sp, spleen; Li, liver. The experiment was performed three times with 3 mice per group. **(d)** Scatter plot indicating numbers of detected Ly6G⁺ cells per high-power field (HPF) of the terminal ileum from untreated BALB/c mice or from BALB/c mice on day 3 after allo-HCT (C57BL/6 $\rightarrow$ BALB/c) or syn-HCT (BALB/c $\rightarrow$ BALB/c) conditioned with either TBI or BU/CY. Data are pooled from two experiments and represent 10 mice per group in total, with 2 HPFs counted per mouse. **(e)** Scatter plot indicating numbers of detected polymorphonuclear cells per HPF of the terminal ileum from untreated BALB/c mice or from BALB/c mice on day 3 after allo-HCT (C57BL/6 $\rightarrow$ BALB/c) or syn-HCT (BALB/c $\rightarrow$ BALB/c) conditioned with either TBI or BU/CY. Data are pooled from two experiments and represent 10 mice per group, with 2 HPFs counted per mouse. An unpaired Student's *t*-test (two-sided) was applied, and the means are shown. **(f)** Representative flow cytometry contour plots of frequency of intestinal neutrophils (CD45⁺, CD11b⁺, F4-80⁺ and Ly6G⁺) in untreated BALB/c mice or from BALB/c mice on day 3 after TBI with 9 Gy or BU/CY. The experiment was performed three times with similar results. $n = 11$ in the untreated and TBI allo-HCT groups, $n = 10$ in the BU/CY and TBI syn-HCT groups. **(g,h)** Quantification of relative numbers of neutrophils (percentages of all leukocytes) **(g)** and absolute numbers of neutrophils **(h)** in mice described in **f**. An unpaired Student's *t*-test (two-sided) was applied, and the means $\pm$ s.e.m. are shown.



polymorphonuclear cells (**Fig. 2e**) in TBI or chemotherapy-conditioned allo-HCT and TBI-conditioned syngeneic HCT (syn-HCT) recipients compared to untreated animals. Further characterization of immune cells in the terminal ileum by flow cytometry revealed that TBI- or chemotherapy-induced tissue damage led to an accumulation of CD11b⁺F4/80⁺Ly6G⁺ neutrophils (**Fig. 2f–h**). These data indicate that neutrophils migrate to and accumulate in the terminal ileum on days 3 and 4 following allo-HCT.

We considered two major events to be the most likely involved in mediating the neutrophil recruitment into the terminal ileum. The first of these was tissue damage following TBI or chemotherapy, and the second was alloreactivity-triggered leukotaxis. When using unconditioned and irradiated *Rag2*; *Il2rg*^{−/−} mice as recipients, we

found that tissue damage due to TBI was sufficient for the influx of myeloperoxidase-positive cells and that transplantation of allogeneic hematopoietic cells was not required (**Supplementary Fig. 1c,d**). In agreement with an infiltration of the gastrointestinal tract by host neutrophils that survived TBI, we found that more than 90% of the neutrophils on day 3 were host derived (**Supplementary Fig. 1e**). The anti-Ly6G antibody serum concentration decreased over time with a half-life of 3.1 d and did not prevent expansion of GFP⁺ neutrophils derived from HoxB8 progenitors or donor bone marrow (BM)-derived Ly6G⁺ neutrophils (**Supplementary Fig. 1f–h**). Chemotherapy- and TBI-based conditioning before syn-HCT led to recruitment of neutrophils into the terminal ileum (**Fig. 2d,e**) without causing signs of histopathological GVHD (**Supplementary Fig. 1i**),

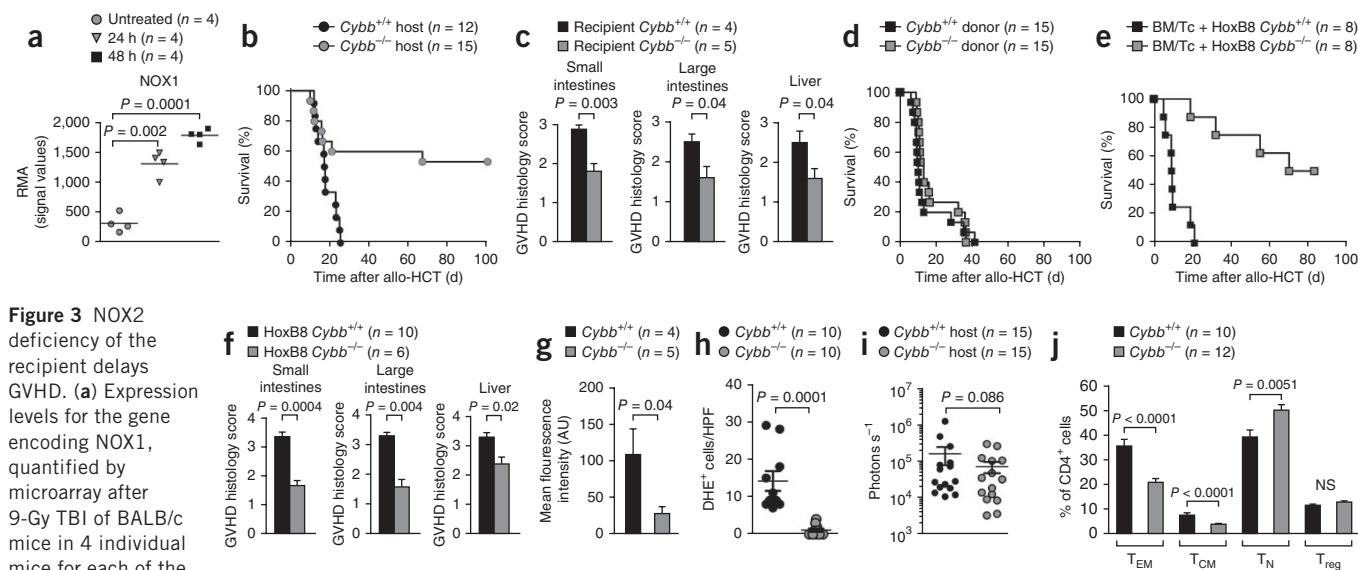


Figure 3 NOX2 deficiency of the recipient delays GVHD. (a) Expression levels for the gene encoding NOX1, quantified by microarray after 9-Gy TBI of BALB/c mice in 4 individual mice for each of the

following groups: untreated, 24 h after TBI and 48 h after TBI. An unpaired Student's *t*-test (two-sided) was applied. (b) Percentage survival of $Cybb^{+/+}$ recipient mice versus $Cybb^{-/-}$ recipient mice (BALB/c→C57BL/6 combination) with a defect in NOX2 ($P = 0.007$; differences in animal survival were analyzed by log-rank test). (c) Histological analysis of the small intestines, large intestines and liver from $Cybb^{+/+}$ and $Cybb^{-/-}$ recipient mice for GVHD severity on day 8. An unpaired Student's *t*-test (two-sided) was used. (d) Survival of WT recipient mice (C57BL/6→BALB/c combination) receiving $Cybb^{+/+}$ versus $Cybb^{-/-}$ BM and T cell grafts (not significant). (e) C57BL/6 recipients (BALB/c→C57BL/6 combination) were BM chimeras with a lack of $Cybb$ in the hematopoietic system (BM/Tc) and received WT or $Cybb^{-/-}$ HoxB8 cells (5×10^6) that were shown to differentiate into neutrophils. WT versus $Cybb^{-/-}$ HoxB8 cells groups, $P = 0.0001$; differences in animal survival were analyzed by log-rank test, and data are pooled from two experiments. (f) Histological analysis of the small intestines, large intestines and liver of $Cybb^{+/+}$ and $Cybb^{-/-}$ HoxB8 mice for GVHD severity on day 8. An unpaired Student's *t*-test (two-sided) was used. (g) Staining for ROS by CellroX shown as mean fluorescence intensity (arbitrary units) in all leukocytes of the peripheral blood for $Cybb^{-/-}$ versus $Cybb^{+/+}$ mice on day 4 after allo-HCT. (h) DHE staining for *in situ* ROS detection in cells in small intestines isolated on day 8 after allo-HCT. An unpaired Student's *t*-test (two-sided) was applied. (i) Proliferation of luciferase transgenic T cells measured by BLI of the abdominal area after allo-HCT. A representative time point (day 8) is shown for 15 animals per group, $Cybb^{+/+}$ versus $Cybb^{-/-}$ recipient mice. An unpaired Student's *t*-test (two-sided) was applied. (j) The frequency of effector memory T cells (T_{EM}), central memory T cells (T_{CM}), naive T cells (T_N) and FoxP3⁺ regulatory T cells (T_{reg}) in the spleen of $Cybb^{+/+}$ or $Cybb^{-/-}$ recipients. An unpaired Student's *t*-test (two-sided) was used. Data are presented as mean $\pm$ s.e.m.

supporting the concept that neutrophils do not induce GVHD by themselves but rather cause tissue damage that promotes T cell activation, which then contributes to GVHD. The majority of the myeloperoxidase-positive cells (>90%) in the ileum after TBI and allo-HCT represented polymorphonuclear neutrophils (Supplementary Fig. 1j).

Microarray-based analysis demonstrated a strong upregulation of the inducible phagocyte NADPH oxidase-1 (NOX1) in the gastrointestinal tract of irradiated compared to untreated mice (Fig. 3a). A pathogenic role for NOX1 had been previously described in chemotherapy-induced intestinal mucositis⁶. Because neutrophils express NOX2 (also known as gp91^{phox}, encoded by *Cybb*), we next investigated whether mice lacking expression of *Cybb*, which is required for generating ROS⁷, had a different GVHD phenotype.

We observed that *Cybb*-deficient allo-HCT recipients had some protection from GVHD (Fig. 3b). Improved survival of $Cybb^{-/-}$ mice correlated with a lower GVHD histopathology score (Fig. 3c). Conversely, using $Cybb^{-/-}$ mice as donors had no effect on survival (Fig. 3d). To specifically determine the importance of *Cybb* in neutrophils, chimeric mice lacking *Cybb* in the hematopoietic system received either wild-type (WT) or $Cybb^{-/-}$ HoxB8 neutrophils^{8,9}. In this model, host-type HoxB8 neutrophil progenitors generated *in vitro* are transferred into allo-HCT recipients. The progenitor cells are expanded *in vitro* using an estrogen-responsive HoxB8 construct^{8,9}. When HoxB8 is turned off by estrogen withdrawal (or upon transferring the cells into mice), progenitor cells differentiate into neutrophils^{8,9}. Ly6G-depleted chimeric mice receiving $Cybb^{-/-}$ neutro-

phils experienced improved survival and reduced GVHD severity compared to mice receiving WT neutrophils (Fig. 3e,f). The number of ROS-producing leukocytes was reduced in $Cybb^{-/-}$ mice compared to $Cybb^{+/+}$ mice on day 4 after allo-HCT (Fig. 3g). Staining of the small intestine on day 8 with dihydroethidium (DHE)¹⁰ revealed an almost complete absence of ROS in $Cybb^{-/-}$ mice undergoing allo-HCT, whereas we found numerous DHE-positive areas in $Cybb^{+/+}$ mice (Fig. 3h). These findings are compatible with previous work showing that the release of ROS contributes to GVHD¹¹.

Recruitment of luciferase-transgenic donor T cells to the intestines in $Cybb^{-/-}$ mice was not impaired (Fig. 3i). However, $Cybb^{-/-}$ mice on day 3 showed a lower frequency of memory T cells in the spleen compared to $Cybb^{+/+}$ mice developing acute GVHD, whereas naive T cells were increased in $Cybb^{-/-}$ mice (Fig. 3j), which is consistent with the previously reported role of memory T cells in acute GVHD¹². Patients with *Cybb* deficiency develop chronic granulomatous disease, but severe GVHD occurs only rarely¹³ or not at all¹⁴.

Because neutrophils are recruited rapidly to sites of bacterial infection, we considered the possibility that neutrophils accumulate during the pathogenesis of GVHD as a result of bacterial leakage into peri-intestinal tissues. To study bacterial translocation, we labeled *Escherichia coli* with supraparamagnetic iron oxide (SPIO) nanoparticles and injected the SPIO⁺ *E. coli* (Supplementary Fig. 2a) into the rectum of mice following TBI or no treatment. When we performed magnetic resonance imaging¹⁵ of the mice 4 h after injection, a strong SPIO signal was still present in the rectum (Supplementary Fig. 2b). After 24 h, the signal derived from SPIO⁺ *E. coli* was gone from the

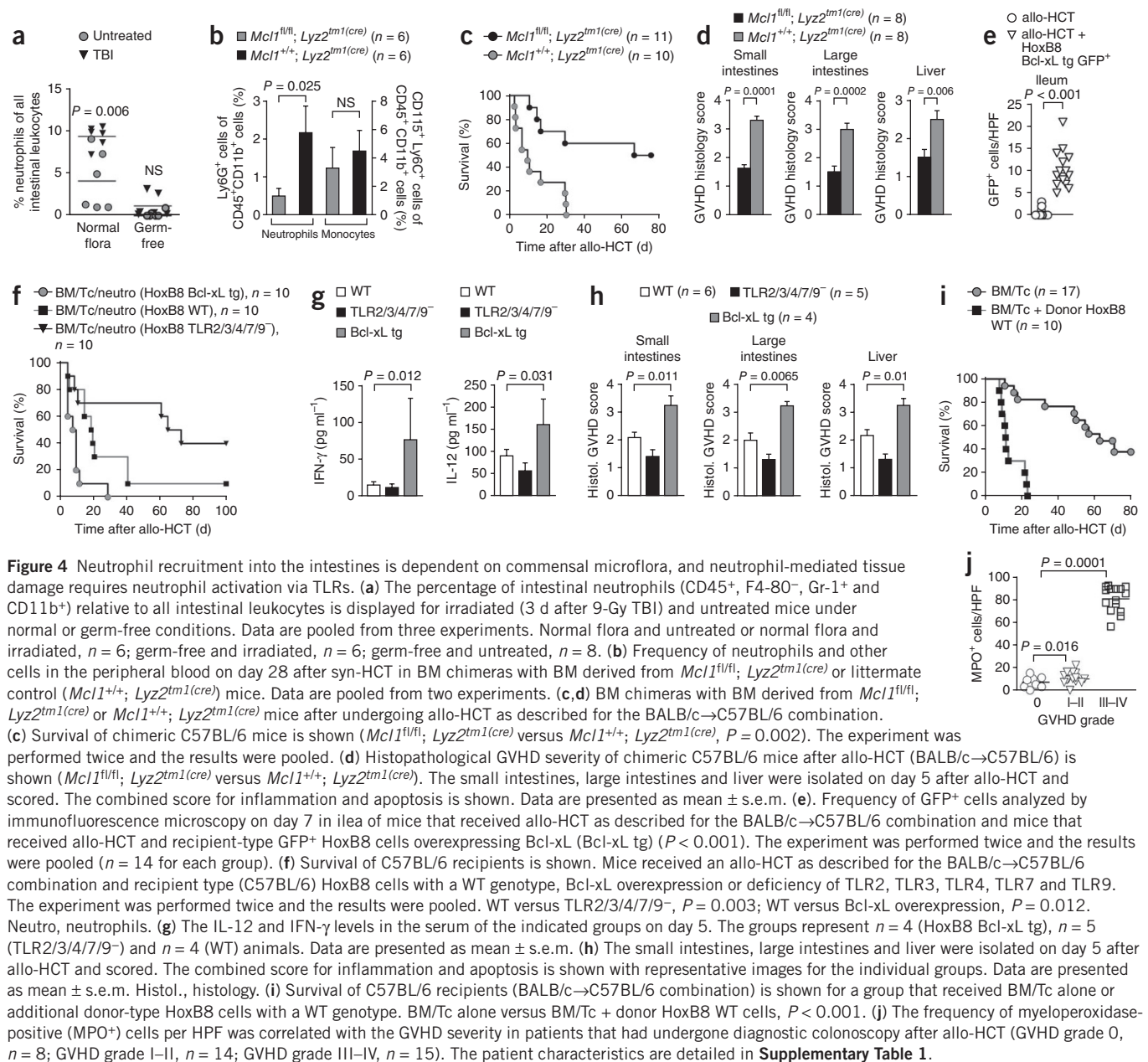


Figure 4 Neutrophil recruitment into the intestines is dependent on commensal microflora, and neutrophil-mediated tissue damage requires neutrophil activation via TLRs. **(a)** The percentage of intestinal neutrophils (CD45⁺, F4-80⁻, Gr-1⁺ and CD11b⁺) relative to all intestinal leukocytes is displayed for irradiated (3 d after 9-Gy TBI) and untreated mice under normal or germ-free conditions. Data are pooled from three experiments. Normal flora and untreated or normal flora and irradiated, $n = 6$; germ-free and irradiated, $n = 6$; germ-free and untreated, $n = 8$. **(b)** Frequency of neutrophils and other cells in the peripheral blood on day 28 after syn-HCT in BM chimeras with BM derived from *Mcl1^{fl/fl}; Lyz2^{tm1(cre)}* or littermate control (*Mcl1^{+/+}; Lyz2^{tm1(cre)}*) mice. Data are pooled from two experiments. **(c,d)** BM chimeras with BM derived from *Mcl1^{fl/fl}; Lyz2^{tm1(cre)}* or *Mcl1^{+/+}; Lyz2^{tm1(cre)}* mice after undergoing allo-HCT as described for the BALB/c→C57BL/6 combination. **(c)** Survival of chimeric C57BL/6 mice is shown (*Mcl1^{fl/fl}; Lyz2^{tm1(cre)}* versus *Mcl1^{+/+}; Lyz2^{tm1(cre)}*, $P = 0.002$). The experiment was performed twice and the results were pooled. **(d)** Histopathological GVHD severity of chimeric C57BL/6 mice after allo-HCT (BALB/c→C57BL/6) is shown (*Mcl1^{fl/fl}; Lyz2^{tm1(cre)}* versus *Mcl1^{+/+}; Lyz2^{tm1(cre)}*). The small intestines, large intestines and liver were isolated on day 5 after allo-HCT and scored. The combined score for inflammation and apoptosis is shown. Data are presented as mean $\pm$ s.e.m. **(e)** Frequency of GFP⁺ cells analyzed by immunofluorescence microscopy on day 7 in ilea of mice that received allo-HCT as described for the BALB/c→C57BL/6 combination and mice that received allo-HCT and recipient-type GFP⁺ HoxB8 cells overexpressing Bcl-xL (Bcl-xL tg) ($P < 0.001$). The experiment was performed twice and the results were pooled ($n = 14$ for each group). **(f)** Survival of C57BL/6 recipients is shown. Mice received an allo-HCT as described for the BALB/c→C57BL/6 combination and recipient type (C57BL/6) HoxB8 cells with a WT genotype, Bcl-xL overexpression or deficiency of TLR2, TLR3, TLR4, TLR7 and TLR9. The experiment was performed twice and the results were pooled. WT versus TLR2/3/4/7/9⁻, $P = 0.003$; WT versus Bcl-xL overexpression, $P = 0.012$. **(g)** The IL-12 and IFN- γ levels in the serum of the indicated groups on day 5. The groups represent $n = 4$ (HoxB8 Bcl-xL tg), $n = 5$ (TLR2/3/4/7/9⁻) and $n = 4$ (WT) animals. Data are presented as mean $\pm$ s.e.m. **(h)** The small intestines, large intestines and liver were isolated on day 5 after allo-HCT and scored. The combined score for inflammation and apoptosis is shown with representative images for the individual groups. Data are presented as mean $\pm$ s.e.m. **(i)** Survival of C57BL/6 recipients (BALB/c→C57BL/6 combination) is shown for a group that received BM/Tc alone or additional donor-type HoxB8 cells with a WT genotype. BM/Tc alone versus BM/Tc + donor HoxB8 WT cells, $P < 0.001$. **(j)** The frequency of myeloperoxidase-positive (MPO⁺) cells per HPF was correlated with the GVHD severity in patients that had undergone diagnostic colonoscopy after allo-HCT (GVHD grade 0, $n = 8$; GVHD grade I–II, $n = 14$; GVHD grade III–IV, $n = 15$). The patient characteristics are detailed in **Supplementary Table 1**.

rectum, and signals from SPIO⁺ *E. coli* in a dipole-shaped pattern had appeared in the tissue surrounding the intestinal wall (**Supplementary Fig. 2c,d**). We detected SPIO in the mesenteric lymph follicles close to the intestinal wall (**Supplementary Fig. 2e**), saw co-localization of neutrophils and bacteria in the intestinal tissue (**Supplementary Fig. 3a–g**) and, consistent with intestinal leakage, detected increased serum lipopolysaccharide (LPS) levels after TBI followed by allo-HCT (**Supplementary Fig. 3h**).

To assess whether the recruitment of neutrophils to the ileum following TBI might be caused by invading bacteria, we compared the frequency of neutrophils in the ilea of mice with normal flora with that in the ilea of mice raised under germ-free conditions. Germ-free mice did not show the increased frequency of neutrophils that we observed in the ilea of mice with normal flora after TBI (**Fig. 4a**), which is compatible with the finding that intestinal inflammation is regulated by the microbiota¹⁶.

The marker Ly6G, used to deplete neutrophils, is also present on other cell types¹⁷. Therefore, to validate the role of neutrophils in GVHD, we generated chimeric mice using BM from *Mcl1^{fl/fl}; Lyz2^{tm1(cre)}* mice, which specifically lack neutrophils¹⁸. We confirmed the selective lack of neutrophils (**Fig. 4b**), and found that *Mcl1^{fl/fl}; Lyz2^{tm1(cre)}* BM chimeras had improved survival (**Fig. 4c**) and reduced histological GVHD severity after allo-HCT (**Fig. 4d**) compared to chimeric *Mcl1^{+/+}; Lyz2^{tm1(cre)}* controls.

To clarify the link between invading bacteria and neutrophil activation, we depleted endogenous neutrophils via injection of anti-Ly6G antibody and then transferred HoxB8 neutrophil progenitors expressing GFP⁹ into the mice (**Supplementary Fig. 1g** and **Fig. 4e**). To assess the contribution of bacterial activation of neutrophils, we compared WT neutrophils to neutrophils lacking TLR2, TLR3, TLR4, TLR7 and TLR9 (TLR2/3/4/7/9⁻). As these cells are unresponsive to most bacterial pathogen-associated molecular patterns, we expected them to be less

activated by translocating bacteria. Another group of mice received HoxB8 neutrophils that overexpressed the antiapoptotic protein Bcl-xL, which prolongs neutrophil survival¹⁹ and can enhance inflammation²⁰.

Transfer of TLR2/3/4/7/9⁻ HoxB8 neutrophils caused less severe GVHD than transfer of WT HoxB8 neutrophils (Fig. 4f). Although deficiency of MyD88, which is used by almost all TLRs (except TLR3) to activate the transcription factor nuclear factor- κ B, in the entire hematopoietic system of the recipient did not protect against GVHD²¹, we observed that a selective deficiency for TLRs in neutrophils did.

Mice receiving HoxB8 neutrophils overexpressing Bcl-xL died more rapidly than the control group receiving WT HoxB8 neutrophils (Fig. 4f), indicating that prolonged survival of neutrophils transgenically overexpressing Bcl-xL enhanced tissue destruction. The reduced survival of mice receiving neutrophils transgenically overexpressing Bcl-xL correlated with higher interleukin-12 (IL-12) and IFN- γ serum levels (Fig. 4g) and higher GVHD histopathology scores (Fig. 4h) compared to the WT group. Although neutrophils in the bloodstream are short-lived, their lifespan can be substantially extended by the presence of IL-1 β in inflamed tissues²². These effects may be relevant for GVHD, as the reported role for IL-1 β in GVHD severity^{23,24} could be via a prolonged survival of neutrophils.

These data indicate that commensal bacteria residing in the gastrointestinal tract cause neutrophil recruitment following TBI. Neutrophils that lack the relevant TLRs for the recognition of bacterial components are associated with less severe GVHD, whereas neutrophils with enhanced resistance to apoptosis due to overexpression of Bcl-xL are associated with more aggressive GVHD.

Transferring donor-type HoxB8 neutrophils caused an increase in GVHD-related mortality comparable to that caused by recipient-type HoxB8 neutrophil transfer (Fig. 4i), indicating that the contribution of neutrophils was independent of MHC expression but dependent on the abundance of the cell population. We prospectively analyzed the number of neutrophils in the gastrointestinal tract of 37 patients undergoing allo-HCT (Supplementary Table 1) and found that the severity of intestinal GVHD strongly correlated with the number of neutrophils present in GVHD lesions (Fig. 4j) and Supplementary Fig. 4a). Staining for nitrotyrosine as a marker of oxidative damage²⁵ correlated with the severity of GVHD (Supplementary Fig. 4b,c).

Our observations that donor-derived neutrophils can contribute to GVHD (Fig. 4i) and that donor neutrophils infiltrate human GVHD lesions (Fig. 4j) are compatible with reported findings that impairment of the LPS-TLR4 axis reduces GVHD severity²⁶ and that high numbers of neutrophils in the gastrointestinal tract are associated with unfavorable transplant outcome²⁷. These findings, along with the correlation of neutrophil numbers in GVHD lesions with clinical GVHD severity, suggest that neutrophils could also play a pathogenic role in development of GVHD in humans. There is accumulating evidence that the gastrointestinal tract is a central location where danger signals are released and abnormal immune responses leading to GVHD^{28,29} are shaped, for example, by the release of ATP serving as a danger signal³⁰ and microbial components released by the intestinal flora¹⁶. Neutrophils recruited into tissues upon bacterial invasion are the first line of defense against microbes. However, in addition to their beneficial bactericidal activity, neutrophil-derived products can induce cellular injury, as shown, for example, in bacterial meningitis²⁰ and inflammatory liver injury³¹.

Overall, our study indicates that TBI and chemotherapy lead to translocation of bacteria into the peri-intestinal tissue, which in turn causes recruitment of neutrophils. Consecutively, the events involve

local activation of neutrophils by TLRs, neutrophil-mediated tissue damage by the release of ROS and donor T cell differentiation into effector cells causing more aggressive GVHD. Therefore, we propose a new proinflammatory role for neutrophils in the pathogenesis of GVHD.

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Accession codes. The complete gene expression data are available at ArrayExpress with accession number [E-MTAB-1566](#).

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

ACKNOWLEDGMENTS

This study was supported by the Deutsche Jose Carreras Leukämie Stiftung (DJCLS grant # R12/11), Deutsche Forschungsgemeinschaft (DFG), Germany, Heisenberg Professorship to R.Z. (DFG ZE 872/3-1) and DFG individual grant to R.Z. (DFG ZE 872/1-2), in part by SFB 850 (to R.Z.), Excellence Initiative of the German Research Foundation (GSC-4, Spemann Graduate School to R.Z. and G.H. and BIOS II, project no. B13 to R.Z.), Deutsche Krebshilfe (to R.Z.), and Marie Curie FP7-IRG268390 (to A.T.) and also in part by the IDEA 2012 award (LSU Health Sciences Center, to G.C.H.). A.M. was funded by the European Research Council and the Wellcome Trust. H.H. was supported by the American Lebanese Syrian Associated Charities (ALSAC). We thank M. Follo for proofreading the manuscript.

AUTHOR CONTRIBUTIONS

L.S., L.G., S.P., F.J.K., S.V.R., A.T., S.G., M.F., F.L., A.S. and K.H. helped design the experiments and performed experiments. S.F.M., R.M., J.F., P.H., M.P., A.M., F.D.v.L. and J.D. helped design the experiments, provided essential reagents and discussed the data. O.S. helped collect patient samples and provided clinical data. H.H. helped design experiments with HoxB8 cells and discussed data. W.R. and N.B. performed and analyzed magnetic resonance imaging experiments. P.W. performed anti-Ly6G antibody measurement in serum of mice, analyzed data and helped write the manuscript. A.S.-G. analyzed GVHD histopathology and neutrophil infiltration. D.P. performed and analyzed microarray experiments. G.C.H., G.H. and R.Z. developed the overall concept, supervised the experiments, discussed the data and wrote the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

Reprints and permissions information is available online at <http://www.nature.com/reprints/index.html>.

1. Tester, A.M. *et al.* LPS responsiveness and neutrophil chemotaxis *in vivo* require PMN MMP-8 activity. *PLoS ONE* **2**, e312 (2007).
2. Sena, L.A. *et al.* Mitochondria are required for antigen-specific T cell activation through reactive oxygen species signaling. *Immunity* **38**, 225–236 (2013).
3. Daley, J.M., Thomay, A.A., Connolly, M.D., Reichner, J.S. & Albina, J.E. Use of Ly6G-specific monoclonal antibody to deplete neutrophils in mice. *J. Leukoc. Biol.* **83**, 64–70 (2007).
4. Giroux, M. *et al.* SMAD3 prevents graft-versus-host disease by restraining T_H1 differentiation and granulocyte-mediated tissue damage. *Blood* **117**, 1734–1744 (2011).
5. Gross, S. *et al.* Bioluminescence imaging of myeloperoxidase activity *in vivo*. *Nat. Med.* **15**, 455–461 (2009).
6. Yasuda, M. *et al.* Potential role of the NADPH oxidase NOX1 in the pathogenesis of 5-fluorouracil-induced intestinal mucositis in mice. *Am. J. Physiol. Gastrointest. Liver Physiol.* **302**, G1133–G1142 (2012).
7. Dale, D.C., Boxer, L. & Liles, W.C. The phagocytes: neutrophils and monocytes. *Blood* **112**, 935–945 (2008).
8. Wang, G.G. *et al.* Quantitative production of macrophages or neutrophils *ex vivo* using conditional Hoxb8. *Nat. Methods* **3**, 287–293 (2006).
9. Gautam, S. *et al.* Survival and differentiation defects contribute to neutropenia in glucose-6-phosphatase- β (G6PC3) deficiency in a model of mouse neutrophil granulocyte differentiation. *Cell Death Differ.* **20**, 1068–1079 (2013).
10. Robinson, K.M. *et al.* Selective fluorescent imaging of superoxide *in vivo* using ethidium-based probes. *Proc. Natl. Acad. Sci. USA* **103**, 15038–15043 (2006).

11. Ziegler, T.R. *et al.* Regulation of glutathione redox status in lung and liver by conditioning regimens and keratinocyte growth factor in murine allogeneic bone marrow transplantation. *Transplantation* **72**, 1354–1362 (2001).
12. Zhang, Y., Joe, G., Hexner, E., Zhu, J. & Emerson, S.G. Alloreactive memory T cells are responsible for the persistence of graft-versus-host disease. *J. Immunol.* **174**, 3051–3058 (2005).
13. Seger, R.A. *et al.* Treatment of chronic granulomatous disease with myeloablative conditioning and an unmodified hemopoietic allograft: a survey of the European experience, 1985–2000. *Blood* **100**, 4344–4350 (2002).
14. Martinez, C.A. *et al.* Excellent survival after sibling or unrelated donor stem cell transplantation for chronic granulomatous disease. *J. Allergy Clin. Immunol.* **129**, 176–183 (2012).
15. Reichardt, W. *et al.* Impact of mammalian target of rapamycin inhibition on lymphoid homing and tolerogenic function of nanoparticle-labeled dendritic cells following allogeneic hematopoietic cell transplantation. *J. Immunol.* **181**, 4770–4779 (2008).
16. Jenq, R.R. *et al.* Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J. Exp. Med.* **209**, 903–911 (2012).
17. Fischer, M.A. *et al.* CD11b⁺, Ly6G⁺ cells produce type I interferon and exhibit tissue protective properties following peripheral virus infection. *PLoS Pathog.* **7**, e1002374 (2011).
18. Dzhagalov, I., St. John, A. & He, Y.W. The antiapoptotic protein Mcl-1 is essential for the survival of neutrophils but not macrophages. *Blood* **109**, 1620–1626 (2007).
19. Kirschnek, S. *et al.* Molecular analysis of neutrophil spontaneous apoptosis reveals a strong role for the pro-apoptotic BH3-only protein Noxa. *Cell Death Differ.* **18**, 1805–1814 (2011).
20. Koedel, U. *et al.* Apoptosis is essential for neutrophil functional shutdown and determines tissue damage in experimental pneumococcal meningitis. *PLoS Pathog.* **5**, e1000461 (2009).
21. Li, H. *et al.* Graft-versus-host disease is independent of innate signaling pathways triggered by pathogens in host hematopoietic cells. *J. Immunol.* **186**, 230–241 (2011).
22. Luo, H.R. & Loison, F. Constitutive neutrophil apoptosis: mechanisms and regulation. *Am. J. Hematol.* **83**, 288–295 (2008).
23. Hill, G.R. *et al.* Differential roles of IL-1 and TNF- α on graft-versus-host disease and graft versus leukemia. *J. Clin. Invest.* **104**, 459–467 (1999).
24. Jankovic, D. *et al.* The Nlrp3-inflammasome regulates acute graft-versus-host disease. *J. Exp. Med.* **210**, 1899–1910 (2013).
25. Forlenza, M. *et al.* Differential contribution of neutrophilic granulocytes and macrophages to nitrosative stress in a host-parasite animal model. *Mol. Immunol.* **45**, 3178–3189 (2008).
26. Cooke, K.R. *et al.* Tumor necrosis factor- α production to lipopolysaccharide stimulation by donor cells predicts the severity of experimental acute graft-versus-host disease. *J. Clin. Invest.* **102**, 1882–1891 (1998).
27. Socié, G. *et al.* Prognostic value of apoptotic cells and infiltrating neutrophils in graft-versus-host disease of the gastrointestinal tract in humans: TNF and Fas expression. *Blood* **103**, 50–57 (2004).
28. Leonhardt, F. *et al.* Inflammatory neovascularization during graft-versus-host disease is regulated by α_v integrin and miR-100. *Blood* **121**, 3307–3318 (2013).
29. Leonhardt, F. *et al.* Spleen tyrosine kinase (Syk) is a potent target for GVHD prevention at different cellular levels. *Leukemia* **26**, 1617–1629 (2012).
30. Wilhelm, K. *et al.* Graft-versus-host disease enhanced by extracellular adenosine triphosphate activating P2X7R. *Nat. Med.* **16**, 1434–1438 (2010).
31. Sitia, G. *et al.* Depletion of neutrophils blocks the recruitment of antigen-nonspecific cells into the liver without affecting the antiviral activity of hepatitis B virus-specific cytotoxic T lymphocytes. *Proc. Natl. Acad. Sci. USA* **99**, 13717–13722 (2002).

ONLINE METHODS

Human subjects. We collected all samples after they were approved for use by the Ethics Committee of the Albert-Ludwigs University Freiburg, Germany (protocol number: 267/11), and after written informed consent was obtained from all patients. In addition, consent to publish was obtained from each patient. Intestinal tissue biopsies were collected in a prospective manner from individuals undergoing allo-HCT with or without GVHD (**Supplementary Table 1**). GVHD grading was performed on the basis of histopathology according to a published staging system^{28,32}. Histochemical staining of the human intestinal biopsies was performed for myeloperoxidase (anti-human MPO, Dako, USA) and chloracetate esterase according to standard protocols.

Mice. C57BL/6 (H-2Kb, Thy-1.2), FVB/N (H-2Kq, Thy-1.2), BALB/c (H-2Kd, Thy-1.2) and BALB/c *Rag2*; *Il2rg*^{-/-} mice were purchased either from Charles River Laboratory (Sulzburg, Germany) or from the local stock of the animal facility at Freiburg University. Experiments done at the Louisiana State University Health Sciences Center, Shreveport, USA used C57BL/6, BALB/c and B6D2F1 mice, which were obtained from The Jackson Laboratory, Bar Harbor, Maine, USA. *Cybb*^{-/-} mice were provided by P.H., and germ-free mice were bred at the facility of the Max Planck Institute of Immunobiology and Epigenetics, Freiburg, Germany. *Mcl1*^{fl/fl}*Lyz2*^{tm1(cre)} and littermate control (*Mcl1*^{+/+}*Lyz2*^{tm1(cre)}) mice lacking neutrophils have been described¹⁸. Mice were used between 6 and 12 weeks of age, and only sex-matched donor-recipient pairs were used. The luciferase (*luc*) transgenic C57BL/6 (Thy1.1) mice have been previously described^{33,34}. Animal protocols were approved by the Federal Ministry for Nature, Environment and Consumer Protection of the state of Baden-Württemberg (protocols G-11/36, G-12/34) or the Animal Committee at Louisiana State University Health Sciences Center-Shreveport (LSUHSC-S), protocol S-11-005 and P-12-033, Shreveport, USA, and the Animal Committee at the University of Utah School of Medicine protocol 13-04002.

Bone marrow transplantation model and histopathology scoring. For bone marrow transplantation experiments, recipients were always injected intravenously with 5×10^6 WT bone marrow (BM) cells directly after lethal TBI. Irradiation was with 9 Gy (split doses of 2×4.5 Gy) for the C57BL/6→BALB/c combination and 10 or 11 Gy (split doses of 2×5.0 or 2×5.5 Gy) for the BALB/c→C57BL/6 and BALB/B→C57BL/6 combinations at Freiburg University, Germany. CD4⁺ and CD8⁺ T cells were isolated from the donor spleen and enriched by positive selection with the MACS cell separation system (Miltenyi Biotec, Germany) using anti-CD4 and anti-CD8 MicroBeads. CD4⁺/CD8⁺ T cell purity was >90% as assessed by flow cytometry (data not shown). The following doses of CD4⁺/CD8⁺ T cells were given intravenously on day 0: 3×10^5 (C57BL/6→BALB/c), 1×10^6 (FVB/N→C57BL/6), 8×10^5 (BALB/c→C57BL/6 WT), 1×10^6 (BALB/B→C57BL/6, minor mismatch), 3×10^5 (BALB/c→BALB/c, syngeneic, Germany) and 1×10^6 (BALB/c→C57BL/6 *Cybb*^{-/-} mice). When indicated, endogenous neutrophils were depleted on day 1 with anti-Ly6G and replaced by HoxB8 cells (C57BL/6), and 1×10^5 T cells were given to induce sublethal GVHD. BALB/c recipients received a 9-Gy single dose followed by i.v. administration of 5×10^6 bone BM cells and 6×10^6 splenocytes of C57BL/6 donors. Using the US syngeneic model B6D2F1→B6D2F1, an irradiation dose of 11 Gy and 6×10^6 splenocytes was used. The different models were employed because the bone marrow transplant experiments were performed either at the University of Freiburg, Germany, or the State University Health Sciences Center Shreveport, USA, and Salt Lake City, Utah, USA, in different animal facilities, with their individual microflora, which in turn required individual irradiation and T cell doses.

Slides of small intestine, large intestine and liver samples collected on days 5–8 were stained with H&E (H/E, Thermo Scientific, USA) and scored by an experienced pathologist blinded to the treatment groups. GVHD severity was determined according to a previously published histopathology scoring system³⁵.

Chemotherapy treatment. Where indicated, mice received busulphan (10 mg/kg/day (d) from d -7 to d -4; Busilvex, Pierre Fabre Pharma, Germany) and cyclophosphamide (100 mg/kg/d from d 3 to d 2; Endoxan, Baxter Healthcare, USA) i.p. (BU/CY) as reported³⁶. For reduced-intensity

conditioning, mice received fludarabine (200 mg/kg/d from d -8 to d -4; Genzyme, Germany) and cyclophosphamide (60 mg/kg/d from d -3 to d -2) i.p. (FLU/CY) as reported³⁷. Day of stem cell injection was assigned as day 0, and the days before stem cell injection are numbered backwards.

Generation of chimeric recipients and transfer of HoxB8 cells. Wild-type (WT) (C57BL/6) recipients were given 5×10^6 *Cybb*^{-/-} or WT BM cells (C57BL/6) or 5×10^6 BM cells derived from *Mcl1*^{fl/fl}*Lyz2*^{tm1(cre)} or littermate control mice (C57BL/6) intravenously after TBI with 9 Gy (2×4.5 Gy). The second HCT (allogeneic) (BALB/c→C57BL/6, T cell dose: 3×10^5) was performed at day 40 after the first HCT (syngeneic). When indicated, the *Cybb*^{-/-} chimeric mice received *Cybb*^{-/-} or WT HoxB8 cells (C57BL/6) (5×10^6 on d 0 of the allo-HCT).

Anti-Ly6G treatment. Anti-Ly6G antibody (Ab) (clone 1A8, Biolegend, Germany, or clone 1A8, BioXCell, USA) or isotype control Ab (Rat IgG2a, κ, Biolegend, Germany, or Rat IgG, Millipore, USA) were injected at the dosage indicated for the individual experiment in the figure legend.

Anti-Ly6G antibody measurement in the serum. For measuring the serum half-life, single doses of 250 μg anti-Ly6G antibody or isotype control antibody were injected i.p. into 5 mice each. Serum samples were collected at different time points between 0 and 15 days after injection. Next, the immunoreactivity of the serum samples was measured by flow cytometry as described previously³⁸. Briefly, 1×10^5 HoxB8-derived neutrophils were incubated with prediluted serum for 1 h on ice. After washing with PBS (Gibco, USA), cells were incubated with goat anti-rat IgG-R-PE secondary antibody (SouthernBiotech, USA) for additional 40 min. Cells were washed again and resuspended in 200 μl PBS containing 1 μg/ml propidium iodide. Mean fluorescence intensities (MFIs) of stained cells were measured and analyzed using a FACScan flow cytometer and the software CellQuest Pro (Becton Dickinson, Germany). Initially, a serum predilution was defined, at which the antibody concentration showed a linear relationship with the MFI. Next, the serum half-life was determined with prediluted sera, setting the initial MFI values of day 0 to 100%.

Subcutaneous injection of complete Freund's adjuvant. In order to model acute inflammation, the inflammatory stimulus complete Freund's adjuvant (CFA) was used. CFA contains dried compounds of *Mycobacterium tuberculosis* emulsified in oil. For induction of acute and local inflammation, CFA was injected s.c. into the footpad of BALB/c WT mice. Injection of CFA was performed with 20 μl emulsion and sterile 500 μl insulin syringes under isoflurane (Abbott, Germany) narcosis.

Bioluminescence imaging. *In vivo* bioluminescence imaging (BLI) for firefly luciferase was performed as previously described³⁹. BLI for the detection of active MPO was performed as follows: 200 mg/kg luminol (Sigma, Germany) were dissolved in distilled water and injected i.p. The mice were imaged in the bioluminescence camera 10 min after injection, with 5 min of luminescent exposure as previously described⁴⁰. *Ex vivo* BLI was performed on different abdominal organs as previously described⁴¹.

E. coli labeling with superparamagnetic iron oxide. In agreement with our previous studies⁴², we have established a procedure to label bacteria (*E. coli*, wild-type, serotype O157:H7) with superparamagnetic iron oxide (SPIO) particles. The *E. coli* bacteria were labeled with a FeraSpin R-Type formulation (nanoPET Pharma GmbH, Germany) consisting of SPIO nanoparticles (average hydrodynamic diameter: 65 nm, core size: 3–5 nm) coated with carboxydextran. SPIO-based contrast agents mainly act on R_2 ($1/T_2$) and R_2^* ($1/T_2^*$) relaxation rates. We exposed *E. coli* to a SPIO concentration of 2 μl/ml for 72 h and 2 consecutive washing steps. The general feasibility of bacterial labeling and tracking with MRI-visible nanoparticles in a high-resolution morphological context has been shown by others⁴³.

Magnetic resonance imaging procedure. Magnetic resonance imaging (MRI) experiments were performed on a conventional horizontal Bruker 7 T Biospec system using a cryogenically cooled quadrature-resonator (Bruker, Germany).

The investigated mice were maintained under anesthesia (2% isoflurane, 98% oxygen) during the MRI procedure. The MRI protocol consisted of a three-dimensional (3D) FLASH sequence (TR/TE = 20/3.04 ms, resolution $50 \times 52 \times 52 \mu\text{m}^3$, flip angle = 10° , bandwidth 55 kHz, acquisition time 9 min 12 s) and a 3D gradient echo (GE) flow compensated sequence (TR/TE = 120/10 ms, flip angle = 22° , spatial resolution $40 \times 40 \times 250 \mu\text{m}^3$, bandwidth 28 kHz, acquisition time 8 min 57 s). Both sequences were chosen to provide morphological information about the migration of SPIO⁺ *E. coli*, based on their R_2 relaxation rates and their magnetic susceptibility, respectively.

For data postprocessing, GE flow compensated images were processed offline with custom-made software developed in Matlab to reconstruct both magnitude and phase images. MR phase images are particularly sensitive to the small deviations from the resonant frequency induced by local perturbations of the magnetic field and hence have significant potential in the visualization of cellular iron-based contrast agent deposits. Specifically, local phase changes are visible due to a difference in susceptibility between the bacteria labeled with SPIO particles and the surrounding tissue. This difference produces a local discontinuity in the magnetic field with high spatial variations over a small distance at the interface between the SPIO particles and the neighboring intestinal tissue. Consequently, a high-pass filter of the phase image will allow suppression of all slowly varying effects and depict only the effects originating from the susceptibility differences⁴⁴. The high-pass filtering operation consists of two steps: (i) a low pass filter of the complex image using a Gaussian filter (equal to the image size) with a normalized $\sigma = 0.14$ and (ii) complex division of the complex image by the low-pass filtered image on a voxel by voxel basis. The resulting image preserves the small-scale phase variations caused by local SPIO, yielding a dipole-shaped pattern on the phase images. Compared to the conventional T_2^* magnitude images, this technique ensures better monitoring of bacterial migration and leakage from the intestines, as well as of bacterial tissue biodistribution.

Neutrophils differentiated from conditional HoxB8-expressing cell lines.

We used a system of conditional HoxB8 expression as previously described⁴⁵ to test molecular and functional defects in neutrophils. Progenitor lines were established that expanded through HoxB8 activity and differentiated into neutrophils when HoxB8 was turned off. We have tested these cell lines for mycoplasma contamination repeatedly and found no contamination. When indicated, WT cells, cells with transgenic expression of Bcl-xL or cells deficient for TLR2, TLR3, TLR4, TLR7 and TLR9 or Cybb were used.

Collection of peripheral vein blood for FACS analysis. For FACS analysis of blood leukocyte subsets, peripheral vein blood was collected from the lateral tail vein in 1 ml ACK lysis buffer (Thermo Scientific, USA) and prepared for further FACS staining.

Flow cytometry. All antibodies were purchased from BD Pharmingen (Germany), BioLegend (Germany), eBioscience (Germany), Abcam (USA), AbD Serotec (Germany) and Jackson IR (USA), and used as FITC, Alexa Fluor 488, PE, PE-Cy5, Pacific blue, Alexa Fluor 647, APC or APC-Cy7 conjugates. Antibodies are described in **Supplementary Table 2**. Data were acquired with a CyanADP (Beckman Coulter; Germany) and then analyzed with FlowJo 7/8 software (Tristar). Cellroxx (Invitrogen, USA) was performed according to the manufacturer's standard protocol. Secondary antibodies were goat anti-mouse IgG2a (FITC, Abcam, GB) and goat anti-rat (Biotin, Jackson IR, USA). All antibodies are listed in **Supplementary Table 2**.

Neutrophil isolation from the intestines for flow cytometry-based analysis and FACS sorting. The procedure of leukocyte isolation from the small intestine after digestion was in principle described before⁴⁶ and applied in this study after slight modifications. Briefly, after removal, the small intestine was cut into 2 pieces of equal size, which were then opened longitudinally in order to wash out feces by cold PBS. To focus on lamina propria leukocytes, the intraepithelial leukocytes and the epithelial layer were removed by use of cell-dissociation buffer (CD buffer; 4% (v/v) FCS, PAN Biotech, Germany, 6% (w/w) collagenase D, Roche, Germany, 6% (w/w) DNase I, Sigma, Germany, in 15 ml (250 U) dispase in HBSS, BD, Germany). Digestion of the tissue was

performed with a shaking frequency of 100 r.p.m. for 20 min at 37 °C three times. Intestinal leukocytes were separated from stromal cells with a Percoll (GE Healthcare, GB) gradient. After centrifugation of the digested intestine, the supernatant was discarded carefully and the pellet was resuspended in 5 ml 40% Percoll. This suspension was carefully loaded on 5 ml 80% Percoll in a 15-ml tube, creating a 40% to 80% gradient with a sharp border. The interphase, containing lamina propria leukocytes, was harvested.

RNA isolation from the small intestines. The terminal ileum was isolated from BALB/c mice that were untreated, at 24 h or 48 h after irradiation. After removal, the small intestine was cut into 2 pieces of equal size, which were then opened longitudinally in order to wash out feces by cold PBS. The tissue was next directly submitted to RNA isolation. The resulting RNA from the terminal ileum of each individual mouse was used to perform a microarray-based transcription rate analysis. RNA was isolated according to a standard protocol (RNeasy mini kit, Qiagen, Germany).

Microarray analysis. RNA samples with an RNA integrity number (RIN) greater than 7 were further processed with the Ambion WT Expression kit (Ambion, USA) as described by the manufacturer. The resulting cDNAs were fragmented and then labeled using the Affymetrix Terminal Labeling kit (Affymetrix, USA). Labeled fragments were hybridized to Affymetrix WT Mouse Gene ST 1.0 arrays for 16 h at 45 °C with 60 r.p.m. in an Affymetrix Hybridization oven 645. After washing and staining, the arrays were scanned with the Affymetrix GeneChip Scanner 3000 7G. CEL files were produced from the raw data with Affymetrix GeneChip Command Console Software Version 3.0.1.

We used the Genedata Expressionist software for further data analysis. CEL files were imported into the Refiner (Version 7.0) module of Expressionist, where preprocessing steps like guanine and cytosine content quantification, background subtraction, quantile normalization and probe summarization were performed⁴⁷. Further analysis was done in the Analyst module (Version 7.0.5a) of Genedata Expressionist, where we ran an analysis of variance with the three experimental groups in order to identify significantly differentially expressed genes. To estimate the false discovery rate, the Benjamini-Hochberg q value was calculated⁴⁸. Only genes from the categories 'main' and 'unmapped' (see Affymetrix transcript annotation NA32) were included, thereby omitting control probes from further analysis. We set our significance thresholds for the effect size to 2-fold change and accepted a false discovery rate of 5% ($q = 0.05$). The complete gene expression data are available at ArrayExpress with accession number [E-MTAB-1566](#).

Conventional histology and immunohistochemistry. Paraffin-embedded sections of 5- μm thickness were mounted on microscope slides (Superfrost/Plus; R. Langenbrink, Germany). Evaluation of tissue sections was performed on a Zeiss Axioplan 2 microscope (Zeiss, Germany). The standard objectives used were 20 $\times$ /numerical aperture 0.45 and 40 $\times$ /numerical aperture 0.60. Digital photos from the microscope were obtained using a Spot digital camera. For immunoenzymatic staining, the tissue was fixed for 10 min in acetone (Sigma, Germany), and the primary antibody was applied (the anti-nitrotyrosine and anti-MPO antibodies are specified in the **Supplementary Table 2**).

Sections were then incubated with a specific secondary biotinylated antibody. Streptavidin horseradish peroxidase and 3,3'-diaminobenzidine were used according to the manufacturer's instructions (Ventana Medical Systems, USA), and sections were counterstained with hematoxylin (Thermo Scientific, USA).

Gram stain for intestinal bacteria. The standard Gram stain procedure for paraffin-embedded tissues was applied as previously described⁴⁹. Briefly, deparaffinized and air-dried tissue sections were incubated for 1 min with crystal violet (Sigma, Germany) staining reagent. After a wash step (2 s), the slide was flooded with Gram's iodine for 1 min followed by a second wash (2 s) and flooding with decolorizing agent. The counterstain was safranin (30 s).

Lipopolysaccharide measurement in the serum. For the measurement of the LPS concentration in the serum, the HEK-Blue LPS Detection Kit (InvivoGen, USA) was used according to the manufacturer's protocol.

HEK-Blue-4 cells were expanded in DMEM, High Glucose (Gibco, USA) with 10% FBS (Hyclone, USA) and 1% penicillin-streptomycin (Gibco, USA).

Cells were selected with HEK-Blue Selection and transferred into HEK-Blue Detection Medium. Blood was obtained from mice by heart puncture and the serum was isolated. 20 μ l of the serum were incubated with 2.5×10^4 HEK-Blue-4 cells. After 18 h on 37 °C in 5% CO₂, the absorbance at a wavelength of 620 nm was measured with an ELISA reader (Tecan, Switzerland). A standard curve was calculated out of the standard row, and the LPS concentration in the serum of the animals was determined.

In situ reactive oxygen species detection. The oxidation of dihydroethidium (DHE; D7008, Sigma, Germany), which is highly specific for *in situ* detection of ROS⁵⁰, was used as a fluorescent assay. Staining of the small intestines was performed on day 8 with DHE according to the manufacturer's recommendation and evaluated on a fluorescence microscope.

Cytokine measurements. The levels of multiple proinflammatory cytokines were analyzed from the serum of mice indicated in the respective experiments with the CBA Inflammation kit (BD, Germany).

Statistical analyses. For the sample size in the mouse GVHD survival experiments, power analysis was performed. A sample size of at least $n = 10$ per group was determined by 80% power to reach statistical significance of 0.05 to detect an effect size of at least 1.06. Differences in animal survival (Kaplan-Meier survival curves) were analyzed by log-rank test.

To obtain unbiased data, the histopathology scoring of GVHD severity was performed by a pathologist blinded to the genotype or treatment group. Only after finalization of the quantitative GVHD severity scores were the samples allocated to their genotypes/treatment group. There was no randomization of mice or samples before analysis. Therefore, all samples or mice were included in our analysis.

For statistical analyses, all data were tested for normality applying the Kolmogorov-Smirnov test. If normality was given, an unpaired *t*-test (two-sided) was applied. If the data did not meet the criteria of normality, the Mann-Whitney *U* test was applied. Data are presented as mean $\pm$ s.e.m. (error bars). Differences were considered significant when the *P* value was <0.05.

32. Lerner, K.G. *et al.* Histopathology of graft-vs.-host reaction (GvHR) in human recipients of marrow from HL-A-matched sibling donors. *Transplant. Proc.* **6**, 367–371 (1974).

33. Leonhardt, F. *et al.* Inflammatory neovascularization during graft-versus-host disease is regulated by α_v integrin and miR-100. *Blood* **121**, 3307–3318 (2013).
34. Zeiser, R. *et al.* Early CD30 signaling is critical for adoptively transferred CD4⁺CD25⁺ regulatory T cells in prevention of acute graft versus host disease. *Blood* **109**, 2225–2233 (2007).
35. Kaplan, D.H. *et al.* Target antigens determine graft-versus-host disease phenotype. *J. Immunol.* **173**, 5467–5475 (2004).
36. Sadeghi, B. *et al.* GVHD after chemotherapy conditioning in allogeneic transplanted mice. *Bone Marrow Transplant.* **42**, 807–818 (2008).
37. Turner, B.E. *et al.* Reduced intensity conditioning for allogeneic hematopoietic stem-cell transplant determines the kinetics of acute graft-versus-host disease. *Transplantation* **86**, 968–976 (2008).
38. Wolf, P. *et al.* Three conformational antibodies specific for different PSMA epitopes are promising diagnostic and therapeutic tools for prostate cancer. *Prostate* **70**, 562–569 (2010).
39. Zeiser, R. *et al.* Differential impact of mTOR inhibition on CD4⁺CD25⁺Foxp3⁺ regulatory T cells as compared to conventional CD4⁺ T cells. *Blood* **111**, 453–462 (2008).
40. Gross, S. *et al.* Bioluminescence imaging of myeloperoxidase activity *in vivo*. *Nat. Med.* **15**, 455–461 (2009).
41. Wilhelm, K. *et al.* Graft-versus-host disease enhanced by extracellular adenosine triphosphate activating P2X7R. *Nat. Med.* **16**, 1434–1438 (2010).
42. Reichardt, W. *et al.* Impact of mammalian target of rapamycin inhibition on lymphoid homing and tolerogenic function of nanoparticle-labeled dendritic cells following allogeneic hematopoietic cell transplantation. *J. Immunol.* **181**, 4770–4779 (2008).
43. Hoerr, V. *et al.* Bacteria tracking by *in vivo* magnetic resonance imaging. *BMC Biol.* **11**, 63 (2013).
44. Haacke, E.M. *et al.* Susceptibility weighted imaging (SWI). *Magn. Reson. Med.* **52**, 612–618 (2004).
45. Wang, G.G. *et al.* Quantitative production of macrophages or neutrophils *ex vivo* using conditional Hoxb8. *Nat. Methods* **3**, 287–293 (2006).
46. Weigmann, B. *et al.* Isolation and subsequent analysis of murine lamina propria mononuclear cells from colonic tissue. *Nat. Protoc.* **2**, 2307–2311 (2007).
47. Irizarry, R.A. *et al.* Exploration, normalization, and summaries of high density oligonucleotide array probe level data. *Biostatistics* **4**, 249–264 (2003).
48. Benjamini, Y. & Hochberg, Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. Series B Stat. Methodol.* **57**, 289–300 (1995).
49. Davies, J.A., Anderson, G.K., Beveridge, T.J. & Clark, H.C. Chemical mechanism of the gram stain and synthesis of a new electron-opaque marker for electron microscopy which replaces the iodine mordant of the stain. *J. Bacteriol.* **156**, 837–845 (1983).
50. Robinson, K.M. *et al.* Selective fluorescent imaging of superoxide *in vivo* using ethidium-based probes. *Proc. Natl. Acad. Sci. USA* **103**, 15038–15043 (2006).